

Macroinvertebrate Measurements

Target grade: Grades 3-5

Time: 45 minutes - 1 hour

Team size: 3-5 students

Introduction:

Students will investigate how scientists use living organisms to evaluate stream health. In this lesson, students will learn what aquatic macroinvertebrates are, the tolerance levels of different macroinvertebrate species, and how to determine water quality based on biodiversity.

Students will work in teams to investigate a model stream. Each team will receive a different stream ecosystem containing a unique combination of insects, crustaceans, amphibians, aquatic plants etc. Students will search their stream, collect the organisms they find, identify and classify them, then use several biological indicators to determine the relative health of their stream.

Each stream will contain a different community of organisms, so students will compare results as a class then engage in a discussion on how the diversity of organisms can provide clues on water quality.

Learning Objectives:

By the end of this lesson students will be able to:

- Define a macroinvertebrate and identify several common freshwater macroinvertebrates
- Distinguish between pollution-sensitive, moderately tolerance, and pollution-tolerant macroinvertebrates
- Analyze stream health using the presence of different organisms
- Support a conclusion about water quality using biological evidence

Materials (Per teams of 3-5 students):

<u>Material</u>	<u>Quantity</u>
Trays	1
Water or blue paper (for model stream)	As needed
Rocks, leaves, plants for stream habitat	As needed
Laminated organism cards or organism cutouts	15-20
Stream organism identification guide	1

Pollution-tolerance reference sheet	1
Student Investigation Worksheet	1
Small try or cup for collecting organisms	1
Tweezers or forceps	1-2

Instructions (Slides):

1. Present slides introducing freshwater stream ecosystems and macroinvertebrates
 - a. Explain what macroinvertebrates are by breaking down the term **macro** and **invertebrate**
2. Show examples of common macroinvertebrates and ask students to potentially identify any, then explain each organism.
3. Introduce the concept of a **bioindicator**
 - a. Explain that a bioindicator is a living organism scientists can study to learn about the condition of an environment
 - b. Explain that different organisms have different levels to tolerance of poor water conditions
4. Introduce the three pollution-tolerance categories and organisms in each
5. Introduce the activity
 - a. Each team will investigate the model stream assigned to them (randomly assign each group Stream 1, 2, or 3 with the respective organisms)
 - b. They will collect all organisms they find
 - c. They will identify and classify each organism
 - d. They will use their observations and evidence to determine the relative water quality of their stream
6. Bring the class back together to:
 - a. Record class results
 - b. Have a discussion with the following questions:
 - i. Which stream appears the healthiest? Why?
 - ii. What differences do you notice between the streams (in terms of macroinvertebrate organisms)?
7. Present the final takeaway slide: What can bugs tell us?
 - a. Review the key concepts learned throughout the lesson:
 - i. Macroinvertebrates are small animals without backbones that are visible to the naked eye.
 - ii. Different macroinvertebrates have different levels of pollution tolerance.
 - iii. Scientists use macroinvertebrates as bioindicators to assess stream health and water quality.
 - iv. Reinforce the main takeaway: Even the smallest organisms can tell us important things about the health of our environment.

Organism Tolerance Classifications Key

Group 1: Sensitive Organisms	Group 2: Moderately Tolerant Organisms	Group 3: Tolerant Organisms
Mayfly nymph	Dragonfly nymph	Aquatic worm
Water penny beetle	Aquatic beetle	Midge larva
Caddisfly larva	Crayfish	Leech

Sample Stream Setups

Stream 1: Healthy Stream	Stream 2: Moderately Impacted Stream	Stream 3: Polluted Stream
4 mayfly nymphs	2 mayfly nymphs	1 caddisfly larva
3 water penny beetles	1 caddisfly larva	1 dragonfly nymph
2 dragonfly nymphs	2 aquatic beetles	5 aquatic worms
1 aquatic beetle	3 snails	5 midge larva
1 crayfish	3 aquatic worms	4 snails
1 frog	1 frog	